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Los Angeles

Forecasting U.S. GDP from 1960 to 2023 using PCA

A thesis submitted in partial satisfaction

of the requirements for the degree

Masters of Science in Statistics

by

Katherine Li

2024

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2024

ABSTRACT OF THE THESIS

Forecasting U.S. GDP from 1960 to 2023 using PCA

by

Katherine Li

Masters of Science in Statistics

University of California, Los Angeles, 2024

Professor George Michailidis, Chair

The U.S. Federal Reserve Board goes to great lengths to make quarterly forecasts of GDP as accurate as possible, as these forecasts play a key role in informing monetary policy that affects millions of Americans. It is therefore not surprising that there is ongoing effort to refine and redefine existing GDP forecasting models so they are more accurate, especially in the event of random shocks to the economy. During the COVID-19 shock, few economic models were able to capture the full volatility of the GDP swings resulting from a once-in-a-lifetime global pandemic. Since then, a whole slew of models, from MIDAS regression to MF-VAR, have been used to forecast the economic impacts of COVID-19 more realistically. In this thesis, I propose a new approach – applying Principal Component Analysis (PCA) to a mix of monthly and quarterly economic variables – to forecast GDP per quarter from 1960 to 2023. To account for the mixed frequency nature of the data, I build PCA forecasts across three data organizations: 1) using only quarterly data, 2) using stacked data, and 3) using imputed data and select the best fit model. I then assess performance of my model against the most widely used times series methods. I find that my economic data is highly sensitive to overfitting and that the PCA model using only quarterly data performs the best at predicting GDP, including capturing the COVID-19 shocks.

The thesis of Katherine Li is approved.

Ying Nian Wu

Frederic Paik Schoenberg

George Michailidis, Committee Chair

University of California, Los Angeles

2024

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Acknowledgements

I want to extend my deep appreciation to Dr. George Michailidis whose instruction and patience has been instrumental for this thesis.

Chapter 1

Introduction

Each quarter, the U.S. Federal Reserve convenes to align on both short and long-term projections for GDP, among other economic variables. The goal is simple: to produce as realistic an outlook of economic activity as possible with some basic assumptions on economic policy. In turn, these discussions will inform monetary policy that affects millions of Americans in the coming years. Given the policy decisions' big impacts, it is vital that forecasts are made accurately.

In the last few decades, there have been several major shocks to the U.S. economy that have challenged the accuracy of economic forecasts. The 1970s saw over a decade of stagflation. Then, in 2008, the housing market crashed, preceding one of the deepest U.S. recessions since World War II. Finally, in recent years, a global pandemic involving the spread of COVID-19 upended life in the U.S, impacting the economy far more than the prior shocks. Unemployment increased, the stock market declined, and, even in its aftermath, work-from-home policies put in place during the crisis still affect the U.S. workforce.

The frequency of these shocks have led to ongoing efforts to improve the accuracy of economic forecasting models. Still, during the COVID-19 pandemic, many forecasters found it difficult to capture the full extent of the GDP swings. Researchers tried new approaches to model the effects of this once-in-a-lifetime global crisis. Foroni, Marcello, and Stevanic (2020) produce forecasts using a mixed frequency MIDAS model. Babii, Ghysels, and Stiaukas (2020) estimate MIDAS regression for nowcasts using LASSO techniques. Schorfheide and Song (2020) apply a mixed frequency-VAR model (MF-VAR) to create their forecasts.

In this thesis, I propose a new method for making more accurate GDP forecasts using Principal Component Analysis (PCA) techniques on monthly and quarterly economic vari-

ables in FRED. I fit a regression model to the PCA components of data up to some time point t and then I use PCA components of data up to future time points greater than t to estimate a future times series for GDP. While this method can reduce data complexity and likelihood of multicollinearity, organizing the mixed-frequency data and selecting for parameters that produce the best fit model can be challenging. I cover these points in more detail in chapter 3.

The rest of this thesis dives deeper into the time series literature as well as my proposed approach. In chapter 2, I provide a review of popular forecasting methods. I discuss time series models ranging from exponential smoothing and ARIMA to newer models such as MIDAS regression and MF-VAR. In chapter 3, I describe the dataset that I am using. I then talk about the methodology, including how the data is organized for the analysis and how the model is selected. In chapter 4, I benchmark the performance of my model against performances of the models discussed in chapter 2 and interpret my results. Lastly, in chapter 5, I conclude.

Chapter 2

Time Series Overview

Over time, many techniques have enriched the time series literature. These techniques range from older yet still commonly used methods such as exponential smoothing and ARIMA to newer methods such as MIDAS regression and mixed-frequency VAR (MF-VAR). The field continues to develop rapidly too, as recent research has been exploring neural network applications to time series data. In this chapter, I provide greater context for some of the more widely used models in time series analysis.

2.1 Exponential Smoothing Models

Exponential smoothing involves making forecasts by taking weighted averages of historical forecasts and actual observations. It was proposed in the 1950s (Holt, 1957; Winters 1960) and is still used as a benchmark for many forecasting methods. Some model formulations for exponential smoothing include the simple exponential smoothing model and the more general ETS model.

2.1.1 Simple Exponential Smoothing

Simple exponential smoothing works best for data with neither seasonality nor trend. The formula for simple exponential smoothing is:

$$\hat{y}_{T+1|T} = \alpha y_T + (1-\alpha)\hat{y}_{T|T-1},$$

where $\hat{y}_{T+1|T}$ is the one-step-ahead smoothed forecast for time $T+1$, y_T is the observed value at time T, and α is the weighting parameter. As α increases, the weight on historical values decreases and recent values have greater impact on the smoothed forecast.

Selection for α can vary. Landi and Johnson (1967) and Trigg and Pitts (1962) discuss selecting α to minimize costs for replenishing inventory. More generally, Gardner (1985) describes parameter selection over a grid search to minimize the ex post MSE. Still, in practice, a range of 0.1 to 0.3 is often used although it may be better to consider a wider range (Gardner, 1985).

The choice for starting value is another consideration to make when fitting a simple exponential smoothing model. Practices vary from using the mean of the data as the starting point to backcasting the data (Gardner, 1985). However, this step in the modeling process may be more up to discretion given the lack of empirical evidence favoring any particular method (Makridakis and Hibon, 1991).

2.1.2 ETS Models

More complex smoothing can be done through other exponential models. In the ETS framework formalized by Hyndman et al. (2002) to select between models, forecasts are made based on error type, seasonality, and trend, with assumptions on if these components are additive or multiplicative. They generalize the 12 exponential methods into the following equations:

$$l_t = \alpha P_t + (1-\alpha)Q_t \tag{1}$$

$$b_t = \beta R_t + (\phi - \beta)b_{t-1} \tag{2}$$

$$s_t = \gamma T_t + (1-\gamma)s_{t-m} \tag{3}$$

where l_t is the series level at time t, b_t is the slope at time t, s_t is the seasonal component at time t, and m represents the number of seasons in a year. P_t, Q_t, R_t, and T_t will vary

depending on the additive or multiplicative structure of the ETS model[1], and α, β, ϕ, and γ are parameters.

Simple exponential smoothing can be represented in this form, assuming no seasonality and no trend[2]. Other commonly used exponential models that can be derived from these equations include the Holt linear method and Holt-Winters' methods. Exponential models outside of these methods are analogous but less often used. As a result, diving deeper into the Holt models can help with understanding how most ETS models work.

The Holt models' assumptions on additivity and multiplicativity inform the relative importance of the 3 generalized equations. In the Holt linear method, the trend is assumed to be additive while seasonality is assumed to be zero. Equations (1) and (2) are therefore more relevant than equation (3) to the model. α and β are weighting parameters, and ϕ is a dampening parameter introduced by Gardner and McKenzie (1985) to flatten the trend line. This dampening parameter is useful for expressing a trend line in which growth is expected to slow over time.

The Holt-Winters' methods assume an additive trend along with either additive or multiplicative seasonality. Both the additive and multiplicative methods use all 3 equations for forecasting, with the multiplicative method tending to show larger seasonal variation as forecast levels increase. The constant γ is another weighting parameter that varies directly with the volatility of the seasonal component. Similar to the Holt linear method, the trend line in both cases can be dampened through ϕ. The same methods for estimating the weighting parameter in simple exponential smoothing can be applied for estimating the parameters in more complex ETS models as well.

[1]See Table A.1 in the Appendix for more details.

[2]Note the similarity between equation (1) and the simple exponential smoothing equation in the previous subsection.

2.2 Single Variable Regression Models

Regression-based time series models are another popular tool for making forecasts of a variable. Unlike exponential smoothing methods, regression-based time series forecasting depends on finding the best fit equation for the current value against past actual values of the variable. Commonly used single variable regression-based models include the ARIMA model and Fourier Analysis regression.

2.2.1 ARIMA Model

Since its first use by Box and Jenkins in the 1970s (Box, 1970), the ARIMA model has become another benchmark of forecasting accuracy. It models the linear relationship of the variable in the current time period with both past values for that variable and independent errors that are assumed to be identically and normally distributed. There are three parts it can decompose into: the autoregressive model, the moving averages model, and the integrative model. This composition can be seen in the following equation:

$$y'_T = c + \phi_1 y'_{T-1} + ... + \phi_p y'_{T-p} + \theta_1 \epsilon_{T-1} + ... + \theta_q \epsilon_{T-q} + \epsilon_T$$

where y'_T is the differenced series at time T, $\phi_1, ..., \phi_p$ are the autoregressive coefficients, ϵ_T is the error at time T, $\theta_1, ..., \theta_q$ are the error coefficients, and c is a constant.

The autoregressive part represents y_T, the variable at time T, solely as a linear combination of its past values. Mirroring the ARIMA representation above, it can be written in the form of:

$$y_T = c + \phi_1 y_{T-1} + ...\phi_p y_{T-p} + \epsilon_T$$

and referred to as an $AR(p)$ model, meaning an autoregressive model of order p. The usual linear regression steps based on minimizing the MSE follow to get the optimal coefficient values. Data stationarity can simplify the model by restricting the range of values the

autoregressive coefficients can take.

The moving averages part uses past forecast errors in a regression-like model for y_T. Again, following the ARIMA representation above and assuming independent errors that are identically and normally distributed, it can be written as:

$$y_T = c + \epsilon_T + \theta_1 \epsilon_{T-1} + \theta_q \epsilon_{T-q}$$

and referred to as an $MA(q)$ model, meaning a moving averages model of order q. Uniqueness in the moving average coefficients occurs by restricting the absolute value of these coefficients to be less than 1. This restriction is important also because it forces the $MA(q)$ to be invertible, meaning it can be written as an $AR(\infty)$ model.

Finally, for non-stationary data, the integrative part can be used to determine the degree of first differencing that makes the time series as close to stationary as possible. This step is important since differencing can simplify the analysis by reducing trend or seasonality effects on the data. After differencing, the mean will become less volatile, making the data more suitable for regression. The backshift operator B is typically used to show differencing, with $By_T = y_{T-1}$. Depending on the level of differencing d, y'_T in the ARIMA representation can be written as $y'_T = (1 - B)^d y_T$.

2.2.2 Fourier Transform Regression

For seasonal data, regression methods based on first converting a time series y_T into its frequency representation through a Fourier transform then regressing on the transformed points can greatly reduce model complexity. The frequency representation identifies the frequencies that are significant to the data, making it possible to leave out the ones that have less impact on the analysis. The Fourier transform works best with data for which the significant frequencies are constant over time.

The Fourier transform of a time series y_T with p cycles in n observations can be repre-

sented by:

$$\omega_p = \sum_{T=0}^{n-1} y_T \ exp(-2\pi ipT/n)$$

where $p = 0, 1, 2, ..., n-1$ and ω_p is a complex Fourier coefficient. Significant frequencies can be determined through a periodogram, which compares the magnitudes of the ω_p. It is more difficult, however, to identify these frequencies without a dominant peak. Baluev (Baluev, 2008) discusses selecting for important frequencies through statistical significance tests.

Once these frequencies are selected for, the modeling process simplifies to regressing only on the selected frequencies. The spectral representation for the time series y_T is:

$$y_T = \sum_{p=1}^{n/2} [\beta_1(\hat{\omega_p})cos(2\pi\hat{\omega_p}T) + \beta_2(\hat{\omega_p})sin(2\pi\hat{\omega_p}T)]$$

where $\beta_1(\hat{\omega_p})$ and $\beta_2(\hat{\omega_p})$ are the regression coefficients at each frequency $\hat{\omega_p}$. The usual linear regression steps apply to obtain the regression coefficient estimates that minimize the MSE.

2.3 Multi-Variable Regression Models

MIDAS regression and VAR are two popular models for performing regressions involving multiple variables.

2.3.1 MIDAS Regression

MIDAS regression represents the variable of interest as a function of past lags of other variables and can model data that is mixed-frequency. For example, it can map the linear relationship between an annual dependent variable against a monthly independent variable. Ghysels et al. (Ghysels et al., 2004) proposed this method to accommodate macroeconomics and finance data, in which it is common to find relevant information that is high-frequency paired with a low-frequency variable of interest.

A simple linear MIDAS regression can be written as follows (Ghysels et al., 2004):

$$Y_t = \beta_0 + \beta_1 B(L^{\frac{1}{m}}) X_{t-1}^{(m)} + \epsilon_t^{(m)},$$

where $B(L^{\frac{1}{m}}) = \sum_{j=0}^{j^{max}} B(j) L^{\frac{j}{m}}$ is a polynomial of length j^{max} in the $L^{\frac{1}{m}}$ operator, $L^{\frac{j}{m}} x_t = x_{t-\frac{j}{m}}$, j is the number of quarterly lags, and m is the number of times faster at which the high-frequency variable is sampled. Essentially, $L^{\frac{j}{m}}$ on x_t produces the value of x_t lagged by $\frac{j}{m}$ periods.

In practice, the high-frequency data is aggregated to have the same frequency as the low-frequency variable of interest before a standard regression model is applied to the data. There is, however, a trade-off between model complexity and loss of high-frequency information. Moreover, the number of parameters to estimate depends on the number of high-frequency lags in the model. Ideally, the number of parameters can be reduced as much as possible while still preserving enough information in the MIDAS regression.

Comparing to same-frequency distributed lag models[3], Ghysels et al. (2004) note that there is no new theory as far as MIDAS is concerned. The parametrization approaches suggested by Judge, Griffith, Hill, Lutkepohl, and Lee (1985) for distributed lag models can also be applied to MIDAS. Similarly, Akaike (1973), with refinement from Schwarz (1978) among others, offers a different approach involving model fitting information criterion for MIDAS model selection.

2.3.2 VAR Models

In vector autoregressive (VAR) models, each variable of a multivariate time series is written as a linear function of past lags of itself and past lags of the other variables. A p-th order VAR, or a VAR(p), is a VAR model with p lags. For example, assuming a dataset with 2

[3] A distributed lag model maps a dependent variable y_t against lags of an independent time series x_t. It is similar to the representation of the ARMA process I discussed in section 2.2.1 but with the x_t regressors replacing the error term lags $\epsilon_{T-1}, ..., \epsilon_{T-q}$ in the MA part. I do not include a subsection for the distributed lag model given that it is just the MIDAS model with $m = 1$.

variables, x_1 and x_2, a VAR(1) model can be written as follows[4]:

$$x_{t,\ 1} = \alpha_1 + \phi_{11} x_{t-1,\ 1} + \phi_{12} x_{t-1,\ 2} + \epsilon_{t,\ 1}$$

$$x_{t,\ 2} = \alpha_2 + \phi_{21} x_{t-1,\ 1} + \phi_{22} x_{t-1,\ 2} + \epsilon_{t,\ 2}$$

where the α_i's are constant coefficients for variable i, the ϕ's are the variable coefficients, and the $\epsilon_{t,\ i}$ are the error terms for variable i. The error terms are assumed to have 0 mean, a positive semi-definite covariance matrix, and no correlations between lags.

Unlike the models I have described so far, VAR models can better accommodate for non-stationary data. In a multivariate OLS model, the intercept and trend are automatically fitted. In Bayesian VAR (BVAR), parameter estimates are typically not affected by non-stationarity although they do depend on the prior that is used[5]. There are many variations of VAR for both approaches.

MF-VAR Model

MF-VAR is a VAR technique that is used to model mixed-frequency data. Schorfheide and Song (2013) developed it using Bayesian estimates with a Minnesota-style prior to make forecasts for quarterly and monthly data. They formulate the model in the following way:

$$x_t = \phi_1 x_{t-1} + ... + \phi_p x_{t-p} + \phi_c + u_t, \quad u_t \sim N(0, \Sigma) \tag{1}$$

where $x_t = [x'_{m,t},\ x'_{q,t}]'$ is a stacked n x 1 vector, $x_{m,t}$ is an n_m x 1 vector of monthly variables and $x_{q,t}$ is an n_q x 1 vector comprising unobserved monthly variables published at the quarterly frequency.

[4]Note that these equations can also be written in matrix form: $x_t = \alpha + A_1 x_{t-1} + A_2 x_{t-2} + \epsilon_t$, where x_t is a 2 x 1 vector with the values of x_1 and x_2 at time t, α is a 2 x 1 vector with the constant coefficients, the A_i are 2 x 2 matrices with the variable coefficients, and ϵ_t is a 2 x 1 vector of error terms.

[5]A prior that is widely used in BVAR models is the Minnesota prior since it simplifies a high-dimensional problem to one of selecting only two parameters.

Rewriting the form of equation (1), Schorfheide and Song (2013) define $z_t = [x'_t, ..., x'_{t-p-1}]'$ and $\phi = [\phi_1, ..., \phi_p, \phi_c]$ and structure it into a VAR(1) model:

$$z_t = F_1(\phi)z_{t-1} + F_c(\phi) + v_t, \quad v_t \sim iid\ N(0, \Omega(\Sigma)) \tag{2}$$

where the first n rows of $F_1(\phi)$, $F_c(\phi)$, and v_t mirror equation (1) and the remaining rows are defined so that the identities $x_{q,t-l} = x_{q,t-l}$ where $l = 1, ..., p-1$ hold. The n x n submatrix of Ω is Σ and all other elements are 0. Equation (2) is the state-transition equation of the MF-VAR.

There are multiple ways to represent unobserved variables, from imputing zeros to setting the measurement error variance to infinity, but Schorfheide and Song (2013) vary the dimension of the vector of observed values as a function of time. With T as the forecast origin and $T_b \leq T$ as the last period corresponding to the last month of the quarter for which all quarterly observations are available, actual observed monthly values $y_{m,t}$ are represented as:

$$y_{m,t} = x_{m,t},\ t = 1, ..., T_b.$$

Meanwhile, in the case of quarterly and monthly data, each quarterly variable $\tilde{y}_{q,t}$ has at least three lags and can be expressed as:

$$\tilde{y}_{q,t} = \frac{1}{3}(x_{q,t} + x_{q,t-1} + x_{q,t-2}) = \Lambda_{qz} z_t.$$

Let $M_{q,t}$ be a selection matrix that equals the identity matrix if t is in the last month of the quarter and it is otherwise empty. For $y_{q,t}$ having dimension n_q in period with observed quarterly average and otherwise 0:

$$y_{q,t} = M_{q,t}\tilde{y}_{q,t} = M_{q,t}\Lambda_{qz} z_t,\ t = 1, ..., T_b.$$

Then, with $M_{m,t}$ as a selection matrix for monthly variables having observed values, $y_{m,t}$

becomes:

$$y_{m,t} = M_{m,t} x_{m,t},\ t = T_b + 1, ..., T.$$

Following this premise, Schorfheide and Song (2013) propose a forecasting process based on Bayesian inference. The starting point is a joint distribution of observed values $Y_{1:T}$, latent states $Z_{0:T}$, and parameters (ϕ, Σ) conditioned on a pre-sample $Y_{-p+1:0}$ to initialize lags. Through a Gibbs sampler, draws are generated from the posterior distribution of $(\phi, \Sigma)|(Z_{0:T}, Y_{-p+1:T})$ and $Z_{0:T}|(\phi, \Sigma, Y_{-p+1:T})$. Based on each draw from the posterior distribution, it is possible to build a forecast trajectory $Z_{T+1:T+H}$ satisfying the state-transition equation (2) and then aggregate these forecasts into quarterly averages.

This chapter covered some of the most widely used time series modeling techniques. For any data analysis, the technique that is used depends on the context, as these methods vary in the types of data inputs they take and in their implementation complexity. In the next chapter, I dive deeper into the dataset I'm using. After, I discuss my proposed PCA regression method, for which I use some of the techniques described in this chapter as a performance benchmark.

Chapter 3

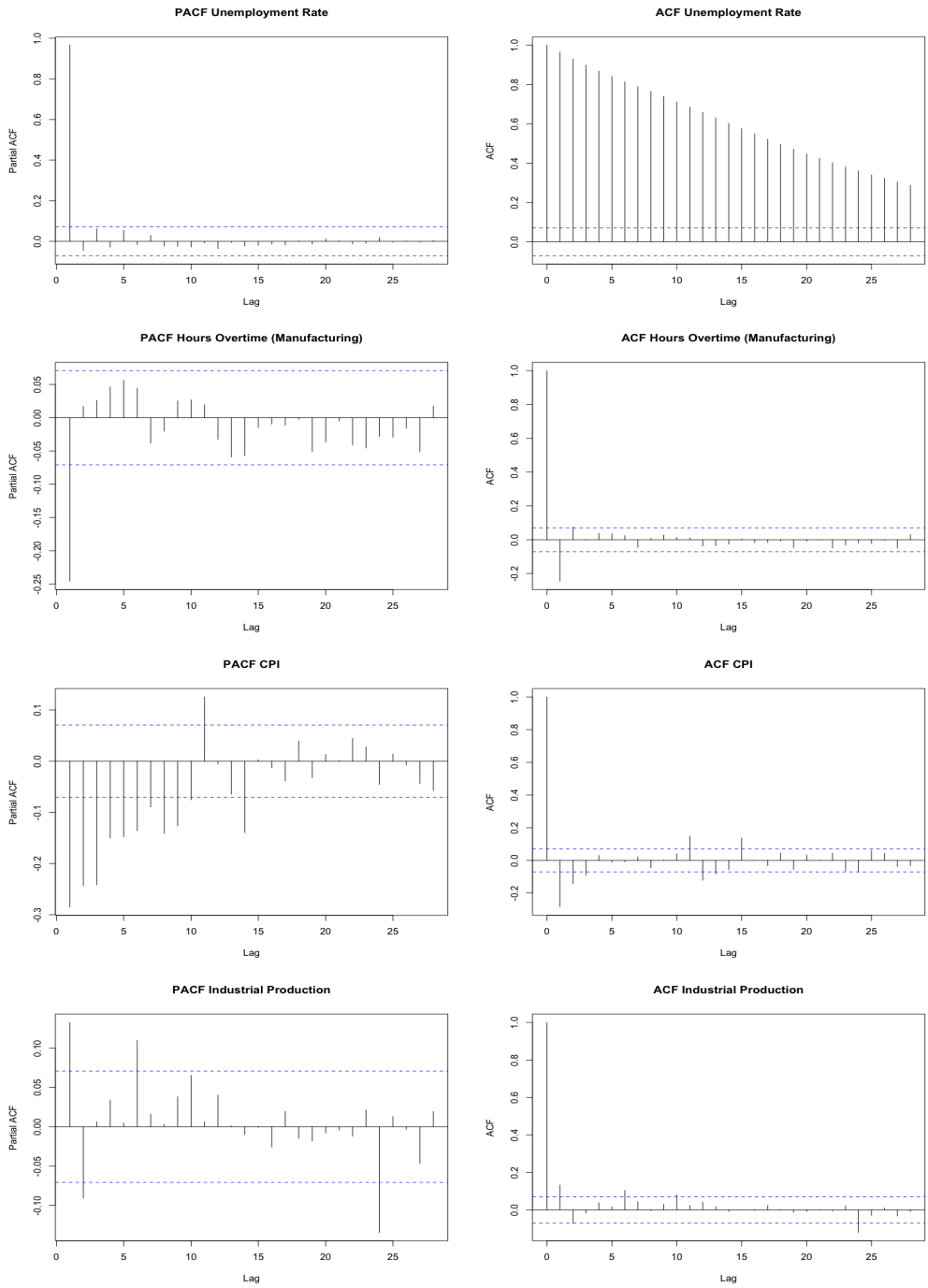
Data and Methodology

3.1 Data Properties

The dataset comes from the Federal Reserve Economic Data (FRED) database and is transformed based on the recommendations of McCracken and Ng (2015, 2016, 2020). It comprises 23 quarterly and 91 monthly U.S. macroeconomic variables. Schorfheide and Song (2020) identify eleven quarterly and monthly variables that are of particular interest. The quarterly variables are GDP, fixed investment, and government expenditure. The monthly variables are unemployment rate, hours worked, Consumer Price Index (CPI), Industrial Production Index (IP), Personal Consumption Expenditure (PCE), Federal Funds Rate, 10-year treasury bond yield, and S&P 500 index. I use these variables as a starting point for my model.

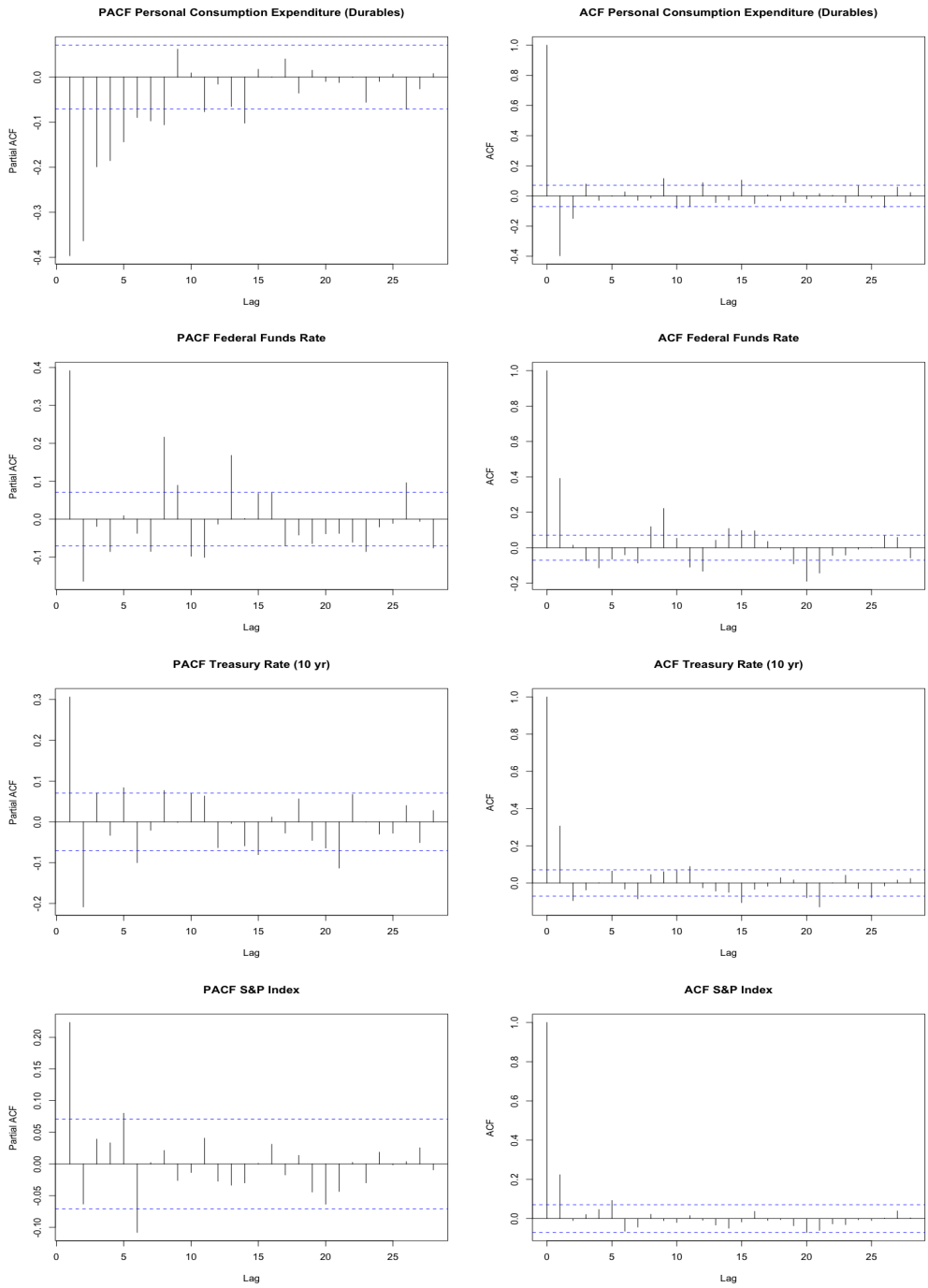
There are a couple of key properties worth noting about the data. GDP is a stationary time series at the 5% significance level by the Augmented Dickey-Fuller (ADF) Test, so many of the techniques discussed in Chapter 2 can be used to make GDP forecasts without further adjustments. The other variables, excluding the unemployment rate, are also stationary at a 5% significance level using the ADF Test. The unemployment rate, although not considered stationary at a 5% significance level, is stationary at a 10% significance level and can be adjusted to be stationary at the 5% significance level through first differencing. By the Kwaitkowski-Phillips-Schmidt-Shin (KPSS) tests, all the time series except GDP, IP, CPI, and the unemployment rate are also stationary at a 5% significance level although restricting GDP, IP, and CPI to the pre-pandemic period would allow them to meet the criteria as well. The unemployment rate failed the criteria for stationarity at the 5% significance level in the

Figure 3.1: PACF and ACF plots for monthly variables (Unemployment rate, Hours Overtime in Manufacturing, CPI, and Industrial Production)



KPSS test, but it passes the criteria after first differencing. In general, since the dataset is relatively stationary, it is possible to apply many time series techniques to it without significant adjustment.

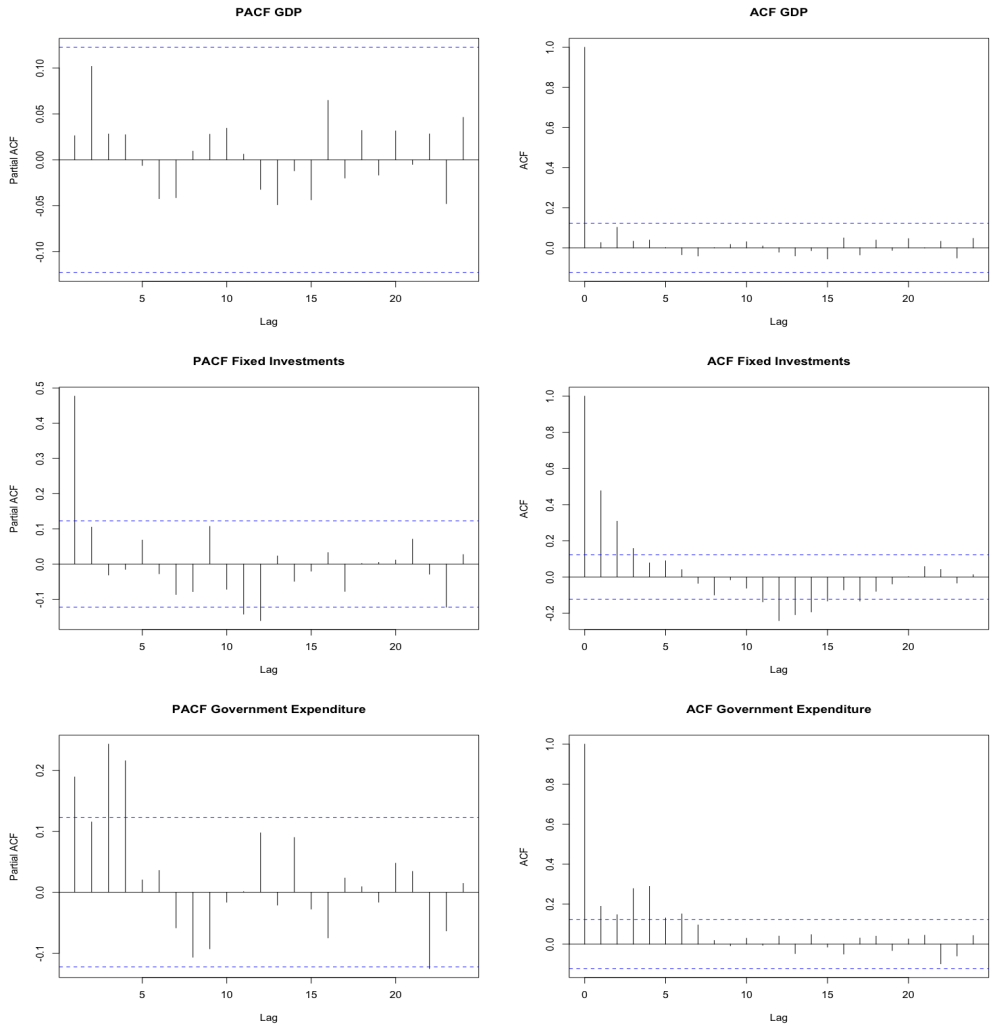
Figure 3.2: PACF and ACF plots for monthly variables (Personal Consumption Expenditure, Federal Funds Rate, 10-yr Treasury Rate, and S&P Index)



Autocorrelations can inform which autoregressive model is the best fit for a time series. For example, to determine the best fit ARIMA(p,d,q) model, I use the ACF plot to select the MA error lag parameter q and the PACF plot to select the AR series lag parameter

p. The GDP time series appears to have no significant autocorrelations in the ACF and PACF plots, making the best fit ARIMA model ARIMA(0,0,0), similar to white noise. The Ljung-Box statistic also shows no significant evidence against the lag autocorrelations being 0. The other time series show more significant correlation, mainly with error terms at 1 time lag. PCE saw significant correlations with error terms up to 2 time lags and CPI saw significant correlations with error terms up to 3 time lags. Hours worked overtime and the 10-year treasury yield had significant autocorrelation and error term correlation, both at 1 time lag.

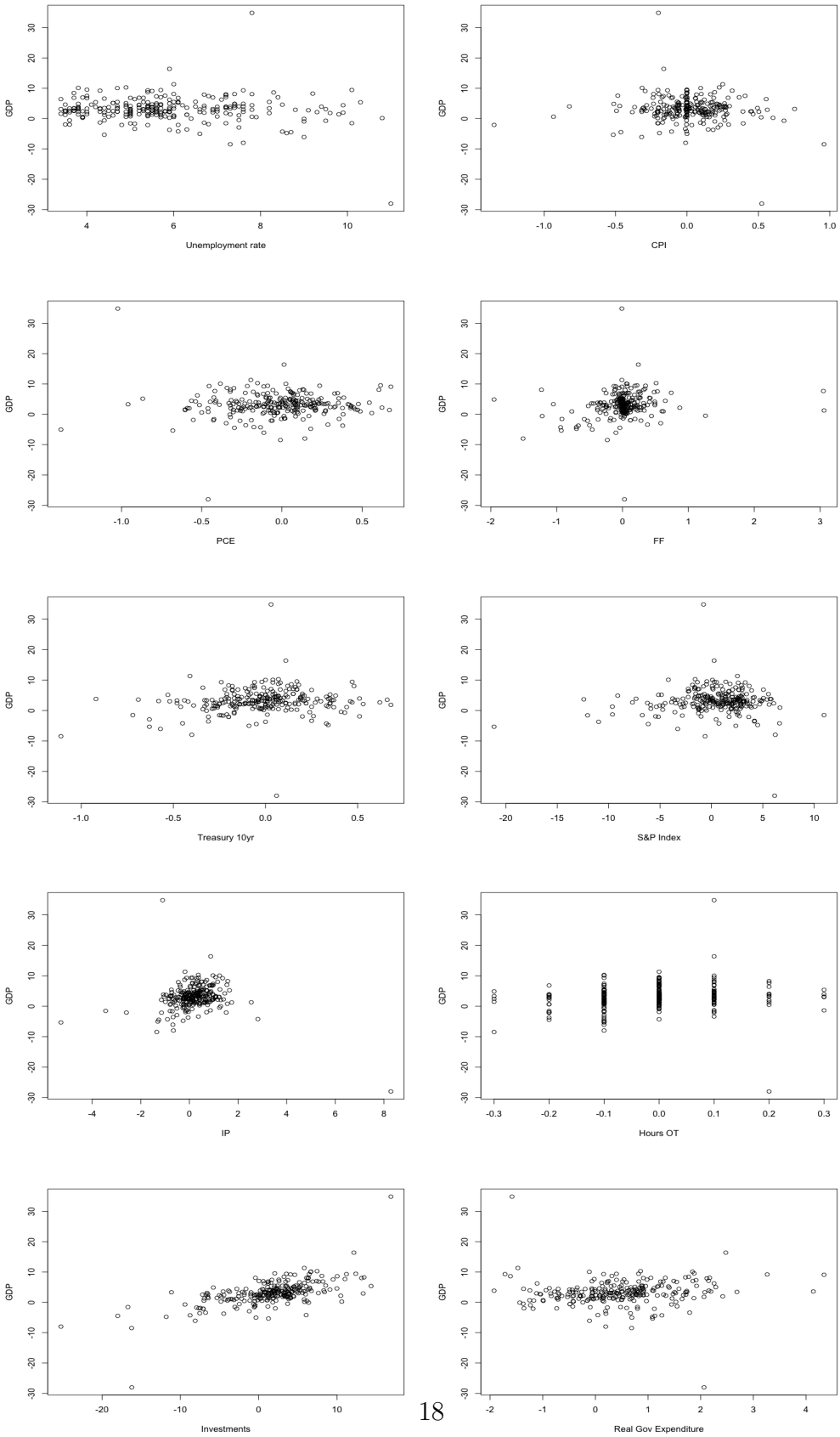
Figure 3.3: PACF and ACF plots for quarterly variables (GDP, Fixed Investments, Government Expenditures)



Among the other quarterly variables, fixed investments saw significant correlation with error terms up to 3 time lags and government expenditure saw significant autocorrelation up to 4 time lags as shown in Figure 3.3. Given the presence of significant autocorrelations, it is possible that there is some residual correlation needing to be adjusted for from regressing GDP onto these other variables.

For most of these 10 other variables of interest, there is no clear linear relationship with GDP. In fact, for monthly variables like unemployment rate, CPI, PCE, the Federal Funds Rate, the 10-year treasury bond yield and the S&P 500 index, as well as for quarterly variables like government expenditures, the absolute value of the correlation coefficient of each variable with GDP is less than 0.2. The graphs for these variables taking only their quarterly values also appear to have no strong relationship with GDP in general. However, the S&P 500 spikes appear to consistently precede GDP spikes by a few time periods. Hours worked overtime in manufacturing has a slightly stronger correlation coefficient of 0.42 but also has no clear relationship with GDP aside from similar quarterly spike and dip patterns during the pandemic. Fixed investment has a stronger positive linear relationship with GDP, with the correlation coefficient being greater than 0.6, and with quarterly spikes and dips being concurrent to or even sometimes preceding GDP spikes and dips. Although industrial production has a low negative linear correlation with GDP, its quarterly spikes and dips also generally co-move or even precede GDP spikes and dips. These initial observations suggest that the S&P 500 index, fixed investments, and industrial production may be good variables for forecasting shifts in GDP. The graphical relationships between GDP and other economic variables are shown in Figure 3.4.

Figure 3.4: GDP versus other economic variables



3.2 Methodology

The methodology that I'm proposing uses PCA to make forecasts. After splitting the data into a training set that spans 1960 to 2012 and a validation set that spans 2013 to 2023, I apply PCA to the training data to simplify it before inputting it into the model. Finally, I linearly regress my variable of interest, GDP, onto the training data principal components to get coefficient estimates. I can then test the regression model's accuracy using the validation data principal components.

3.2.1 Principal Component Analysis (PCA) Overview

PCA linearly transforms data into a new coordinate space to make it easier to identify the dimensions (or principal components) that capture the greatest data variation. The process starts with finding the component of maximum variance for the data by calculating the variance-covariance matrix for some normalized data matrix $\mathbf{X} = (X_1, X_2, ..., X_p)$. Assuming n is the number of training samples, the variance-covariance matrix $\text{var}(\mathbf{X}) = \Sigma = \frac{1}{n}(XX^T)$, for which each element $\sigma_{i,j}$ is the covariance between variables X_i and X_j for $i, j = 1, ..., p$. The first component Y_1 can be written as a linear combination of the data, where, for regression coefficients $e_{11}, ..., e_{1p}$,

$$Y_1 = e_{11}X_1 + e_{12}X_2 + ... + e_{1p}X_p$$

and the data variation can be maximized by selecting the $e_{11}, ..., e_{1p}$ that maximizes

$$var(Y_1) = \Sigma_{k=1}^{p}\Sigma_{l=1}^{p} e_{1k}e_{1l}\sigma_{kl}$$

following the constraint that

$$\Sigma_{j=1}^{p} e_{1j}^2 = 1$$

to guarantee uniqueness of the coefficients.

The next component Y_2 should cover the second most data variance, independent of the first component. The process is the same as with Y_1, but with the added constraint that Y_1 and Y_2 are uncorrelated and

$$cov(Y_1, Y_2) = \Sigma_{k=1}^{p} \Sigma_{l=1}^{p} e_{1k} e_{2l} \sigma_{kl} = 0.$$

Each component Y_r that follows satisfies the same properties, constrained by Y_r being pairwise uncorrelated with the components that precede it.

The constraints on these principal components force the regression coefficients to be orthonormal. A result that satisfies this requirement can be found using the Spectral Decomposition Theorem, which states that for any square matrix A there exists an orthogonal matrix Q and a diagonal matrix D such that $A = QDQ^T$. Therefore, by the Spectral Decomposition Theorem, there exists some orthogonal matrix Q and diagonal matrix D such that the square variance-covariance matrix $\Sigma = XX^T = QDQ^T$, and the diagonal elements of D are the eigenvalues of Σ. With Q being the regression coefficients, the principal components $Y_i = XQ_{i\cdot}$ and $var(Y_i) = Q_{i\cdot}^T var(X) Q_{i\cdot} \propto Q_{i\cdot}^T XX^T Q_{i\cdot} = D_{ii}$, where $i = 1, ..., n$. Since the eigenvalues along the diagonal of the n x n matrix D are ordered from the greatest value D_{11} to the least D_{nn}, the eigenvectors in Q are already arranged so that Y_1 covers the greatest data variance and each proceeding principal component covers less variance.

The number of principal components used can vary per analysis. One common method for making this choice is the scree plot, a graph of variance against principal components. All the principal components before the "elbow" in the scree plot are used, as the variance drop-off after the "elbow" is steep. Another common method for selecting principal components relies on choosing the number of principal components needed to achieve a desired cumulative variation. This threshold cumulative variation can also vary per analysis and is largely up to discretion.

3.2.2 PCA Time Series Model

Suppose the data matrix $X = (X_1, X_2, ..., X_p)$ is an n x p matrix, where n is the number of time periods and p is the number of variables. Let n' be the time period such that any time $t \leq n'$ belongs to the training set X', an n' x p matrix, and any $t > n'$ is in the validation set X'', an $(n - n')$ x p matrix. The data is split non-randomly given that time series forecasting is by nature forward-looking.

By applying PCA to X' after normalizing it, the data can be simplified further before modeling. The variance-covariance matrix for X' is

$$\Sigma = X'X'^T \tag{1}$$

an n' x n' matrix. By the Spectral Decomposition Theorem, there exists an orthogonal matrix Q and diagonal matrix D such that

$$\Sigma = QDQ^T \tag{2}$$

where Q is a matrix of eigenvectors of Σ and D is a matrix of its eigenvalues. The dimension of matrix Q can be reduced so that it contains only as many eigenvectors as satisfies a pre-chosen cumulative variance.

Suppose $\tilde{Q}$ is the n' x $\tilde{n}$ dimension-reduced version of Q, where $\tilde{n} < n'$, and $\tilde{D}$ is its corresponding diagonal matrix of eigenvalues. Then, the dimension-reduced principal components are calculated by

$$\tilde{C} = X'\tilde{Q} \tag{3}$$

where $\tilde{C}$ is an n' x $\tilde{n}$ matrix of $\tilde{n}$ eigenvectors that are each of dimension n' x 1.

This adjusted matrix $\tilde{C}$ is used as input into a VAR regression model where GDP is the dependent variable. I can then cross-validate this model across lag parameter values to select the one that best fits the data based on the validation mean errors. The final model

uses the significant coefficients that produce the lowest error.

Specifically, in order to mimic the process of realizing actual data until the latest period only, the principal components for the validation set are built each period on all values, including training data, up to that time period. Assuming the current time period is $T > n'$, the forecast for GDP in time period $T + 1$ will depend on principal components of the T x p matrix X_T[1]. Then, the relevant component values are entered into each model to produce a one-step ahead forecast of GDP until all periods covered by the validation set have GDP estimates. Once these future forecasts for GDP are constructed, the optimal model can be selected depending on the root mean squared error (RMSE):

$$RMSE = \sqrt{\frac{1}{n - n' - 1} \Sigma_{t=n'+1}^{n} (y_t - \hat{y}_t)^2}$$

where y_t is the GDP and $\hat{y}_t$ is the model's GDP forecast both at time t.

3.3 Mixed-Frequency Data Schemes

I explore three approaches to adjusting data that is sampled at different frequencies. The first approach is to align all data to the lowest frequency. For example, with GDP being a quarterly variable and unemployment rate being a monthly variable, the new data would include only the quarterly values of GDP and the unemployment rate. Another method is to create a stacked vector of quarterly and monthly variables. The last approach is to impute monthly values for the quarterly variables and then model the augmented monthly data. There are different advantages to using each method.

3.3.1 Lowest frequency approach

Aligning all data to the lowest frequency is the simplest approach to structuring the data, although, as Ghysels (2016) points out, it may mis-specify the co-movements between high-

[1]This matrix takes the data in X up to row T.

frequency and low-frequency variables. I start with the full training data X' and apply PCA to its end-of-quarter values only to obtain principal axes. Then, after mapping the data back to enough principal axes that can capture a specific threshold of data variance, I get the $\lfloor \frac{n'}{4} \rfloor$ x $\tilde{n}$ component matrix $\tilde{C}$. The n' x $\tilde{n}$ component matrix

$$\tilde{C} = \begin{pmatrix} \alpha_{1,v_1} & \alpha_{1,v_2} & \dots & \alpha_{1,v_{\tilde{n}}} \\ \alpha_{2,v_1} & \alpha_{2,v_2} & \dots & \alpha_{2,v_{\tilde{n}}} \\ . \\ . \\ . \\ \alpha_{n',v_1} & \alpha_{n',v_2} & \dots & \alpha_{n',v_{\tilde{n}}} \end{pmatrix} \tag{4}$$

becomes

$$\tilde{C}' = \begin{pmatrix} \alpha_{1,v_1} & \alpha_{1,v_2} & \dots & \alpha_{1,v_{\tilde{n}}} \\ \alpha_{4,v_1} & \alpha_{4,v_2} & \dots & \alpha_{4,v_{\tilde{n}}} \\ . \\ \alpha_{\tau,v_1} & \alpha_{\tau,v_2} & \dots & \alpha_{\tau,v_{\tilde{n}}} \\ . \\ \alpha_{\lfloor \frac{n'}{4} \rfloor,v_1} & \alpha_{\lfloor \frac{n'}{4} \rfloor,v_2} & \dots & \alpha_{\lfloor \frac{n'}{4} \rfloor,v_{\tilde{n}}} \end{pmatrix} \tag{5}$$

where the v_i's are the data variables, $\alpha_{.,.}$ is a principal component value, and τ is any time period with a quarterly GDP value. Then, the vector y of GDP values

$$y = \begin{pmatrix} y_1 \\ y_4 \\ . \\ y_\tau \\ . \\ y_{\lfloor \frac{n'}{4} \rfloor} \end{pmatrix}$$

is regressed on $\tilde{C'}$ as shown in equation 5 across multiple lag values to get coefficient estimates $\hat{\beta_{t_4}}$, where $y = \hat{\beta_{t_4}}\tilde{C'}$.

3.3.2 Stacked vector approach

The stacked vector approach was proposed by Ghysels (2016) to overcome the potential misspecification of co-movements between high-frequency and low-frequency variables. Instead of removing monthly values that do not align to end-of-quarter dates, the stacked matrix $\tilde{C_{st}}$ stacks the principal component monthly values with the GDP quarterly values after applying PCA to X'. For example, a monthly component will have values for January, February, and March each while GDP will only have a value for January, all of which will be collected into a single vector. The n' x $((\tilde{n}_H * m) + \tilde{n}_L)$ data matrix, with $\tilde{n}_H$ high frequency variables and $\tilde{n}_L$ low frequency variables in the training set, becomes

$$\tilde{C_{st}} = \begin{pmatrix} \alpha_{v_{H1},1}(\tau_1) & ... & \alpha_{v_{Hn},1}(\tau_1) & ... & \alpha_{v_{H1},m}(\tau_1) & ... & \alpha_{v_{Hn},m}(\tau_1) & \alpha_{v_{L1}}(\tau_0) \\ \alpha_{v_{H1},1}(\tau_2) & ... & \alpha_{v_{Hn},1}(\tau_2) & ... & \alpha_{v_{H1},m}(\tau_2) & ... & \alpha_{v_{Hn},m}(\tau_2) & \alpha_{v_{L1}}(\tau_1) \\ . & & & & & & & \\ \alpha_{v_{H1},1}(\tau_L) & ... & \alpha_{v_{Hn},1}(\tau_L) & ... & \alpha_{v_{H1},m}(\tau_L) & ... & \alpha_{v_{Hn},m}(\tau_L) & \alpha_{v_{L1}}(\tau_{L-1}) \\ . & & & & & & & \\ \alpha_{v_{H1},1}(\tau_{n'}) & ... & \alpha_{v_{Hn},1}(\tau_{n'}) & ... & \alpha_{v_{H1},m}(\tau_{n'}) & ... & \alpha_{v_{Hn},m}(\tau_{n'}) & \alpha_{v_{L1}}(\tau_{n'-1}) \end{pmatrix} \quad (6)$$

where $v_{H\cdot}$ is a high-frequency variable, $v_{L\cdot}$ is a low-frequency variable and τ_i is the quarterly time period. The $\alpha_{v_{H\cdot},m}$ are the values for variable $v_{H\cdot}$ at month m of the quarter that ends with time period τ_i. The low-frequency variable comes from the quarter before so that the latest values for each variable are used in the forecast of the low-frequency variable in the next quarter.

Although the stacked matrix $\tilde{C_{st}}$ can better capture the dynamics between high frequency and low frequency variables, it can also greatly augment the size of the model depending on how many lags are included. To reduce the number of parameters that need to be estimated

and control for overfitting, I take an average of the high frequency variables at any quarterly time period τ_L. As a result, the original stacked matrix $\tilde{C}_{st}$ is modified so that

$$\tilde{C}_{st_{agg}} = \begin{pmatrix} \overline{\alpha}_{v_{H1},1:m}(\tau_1) & \dots & \overline{\alpha}_{v_{Hn},1:m}(\tau_1) & \dots & \alpha_{v_{L1}}(\tau_0) \\ \overline{\alpha}_{v_{H1},1:m}(\tau_2) & \dots & \overline{\alpha}_{v_{Hn},1:m}(\tau_2) & \dots & \alpha_{v_{L1}}(\tau_1) \\ . & & & & \\ \overline{\alpha}_{v_{H1},1:m}(\tau_L) & \dots & \overline{\alpha}_{v_{Hn},1:m}(\tau_L) & \dots & \alpha_{v_{L1}}(\tau_{L-1}) \\ . & & & & \\ \overline{\alpha}_{v_{H1},1:m}(\tau_{n'}) & \dots & \overline{\alpha}_{v_{Hn},1:m}(\tau_{n'}) & \dots & \alpha_{v_{L1}}(\tau_{n'-1}) \end{pmatrix} \tag{7}$$

where $\overline{\alpha}_{v_{H1},1:m}(\tau_L) = \frac{1}{m}\Sigma_{i=1}^{m}\alpha_{v_{H1},i}(\tau_L)$. Bai et al. (2013) show that using ad hoc linear time averaging matches a steady state Kalman filter prediction equation based on a single factor state space model and is a good approximation for more complex state space model specifications.

Lastly, the ordering of the stacked vectors can be random in this analysis although there are cases in which the stacked vectors must follow a specific order. Ghysels (2016) notes that stacking the vector in order of when data is released is important to a real-time analysis but it is not required to analyze the economic time structural dynamics, as in this case. I use the convention of stacking the variables with high frequency variables coming before low frequency variables.

3.3.3 Imputed values approach

The final data scheme comprises imputing values for GDP to match the number of observations for the high-frequency variables. Exponential smoothing can be used to produce a one-step ahead forecast for GDP that fills in its "missing" monthly values. Although this method preserves the time order of the GDP series in the training data, it is difficult to get accurate estimates for monthly GDP especially at the start of the series, when data is very sparse. An alternative method is to use cubic spline interpolation to fill in the monthly gaps

in the training data before using the model to make one-step ahead forecasts for GDP with the validation data.

Given that this approach produces a high-frequency series of GDP values, the regression model is based on all monthly variables. Therefore, after using PCA to simplify the training data, no adjustment for mixed-frequency data is needed to regress GDP on the principal components. The component matrix takes the form of $\tilde{C}$ shown in equation 4. Once the significant regression coefficients are found, they can be applied to the validation data to predict GDP in future quarters.

While this approach makes use of the full dataset, its main limitation is that the monthly imputations introduce biases themselves. Estimating a model on this imputed data can compound the estimation error even more.

Chapter 4

Results and Discussion

4.1 Results

I fit my PCA model to the FRED data under each data scheme and I benchmark its performance against the time series models discussed in chapter 2. In each iteration, it is clear that: 1. using variables outside of GDP lag terms can increase GDP forecast accuracy and 2. the data is highly sensitive to overfitting. In the end, the model producing the lowest mean error is the PCA model applied to data at the lowest frequency.

4.1.1 Benchmark models

I use the ETS model, the ARIMA regression model, and the Fourier transform regression model, which are common benchmarks for time series model performance, to forecast GDP. Forecasting accuracy from applying each of these models to the FRED data is similar, but the ARIMA model slightly outperforms any single model.

ETS Performance

Using Hyndman et al's (2002) model selection criteria on GDP training data spanning 1960 to the end of 2012, I find that the ETS model that minimizes the validation error is the simple exponential smoothing model. This result is not surprising, given that GDP is mostly stationary, so fitting seasonality or a trend line onto the series through smoothing could over-index certain time periods. However, including these non-stationary elements in the model does not change the mean error materially either, likely because the effects from seasonality

or a trend line are generally weak. Table 4.1 shows the RMSE across multiple ETS models. The RMSEs for both the Holt model, which assumes additive trend but no seasonality, and the Holt-Winters model, which assumes both additive trend and additive seasonality, are barely greater than the RMSE for the simple exponential smoothing model.

ARIMA Performance

The best fit ARIMA model for the GDP training data is an ARIMA(1,0,2) model, as the PACF shows a lag correlation of 2 and the ACF shows a lag correlation of 1. The validation RMSE may be slightly less than the validation RMSEs in the ETS models because the ARIMA model also accounts for correlations between GDP lag terms. Hyndman et al. (2008) come up with the same best fit ARIMA model based on minimizing the AIC and BIC values.

A slight variation on the ARIMA model adds a spectral component that captures the dominant frequencies in the GDP training data. This model outperforms the ETS and Fourier transform regression models but is slightly worse than the ARIMA model alone. The outcome is as expected, given that the data is not super seasonal but the Fourier analysis is more suitable for very seasonal data. I use two frequency components in the Fourier summand, which I select through cross validation.

Fourier transform regression performance

Table 4.1 shows that, of the benchmark models, the Fourier transform regression by itself has the highest validation RMSE. This result is expected given that GDP is generally a stationary time series while Fourier analysis is more suited toward modeling time series with strong seasonality. The Fourier transform for GDP in fact identifies seven dominant frequencies for GDP, of which only three had significant coefficients after regressing GDP onto them. Only one of these frequencies was also used in the Fourier terms of the hybrid ARIMA-Fourier transform model discussed in the previous subsection.

Table 4.1: This table shows the training and validation RMSEs for the benchmark and PCA models on FRED data. The training RMSE is lower for nearly every PCA model than the benchmark model. The validation error is similar across benchmark models and lower in specific PCA models.

	Training RMSE	**Validation RMSE**
Benchmark Models		
ETS optimal (SES)	3.46	7.12
ETS Holt	3.47	7.14
ETS Holt-Winters	3.44	7.16
ARIMA	3.28	7.05
Fourier Regression	3.28	7.31
ARIMA + Fourier	3.23	7.08
PCA 1-lag models		
Lowest frequency (full data)	2.66	3.17
Lowest frequency (S&S data)	2.73	11.14
Stacked vector (full data)	2.15	5.57
Stacked vector (S&S data)	2.59	9.68
Stacked averages vector (full data)	2.32	10.03
Stacked averages vector (S&S data)	3.31	8.34
Imputed spline (full data)	1.95	8.81
Imputed spline (S&S data)	1.99	6.11

4.1.2 PCA model across different data schemes

Performance of the PCA model varied across data schemes. Applying the PCA model to data adjusted to the lowest frequency performed the best while imputing the data before applying any model performed the worst. The data is highly sensitive to overfitting, which may affect model performance in each data scheme.

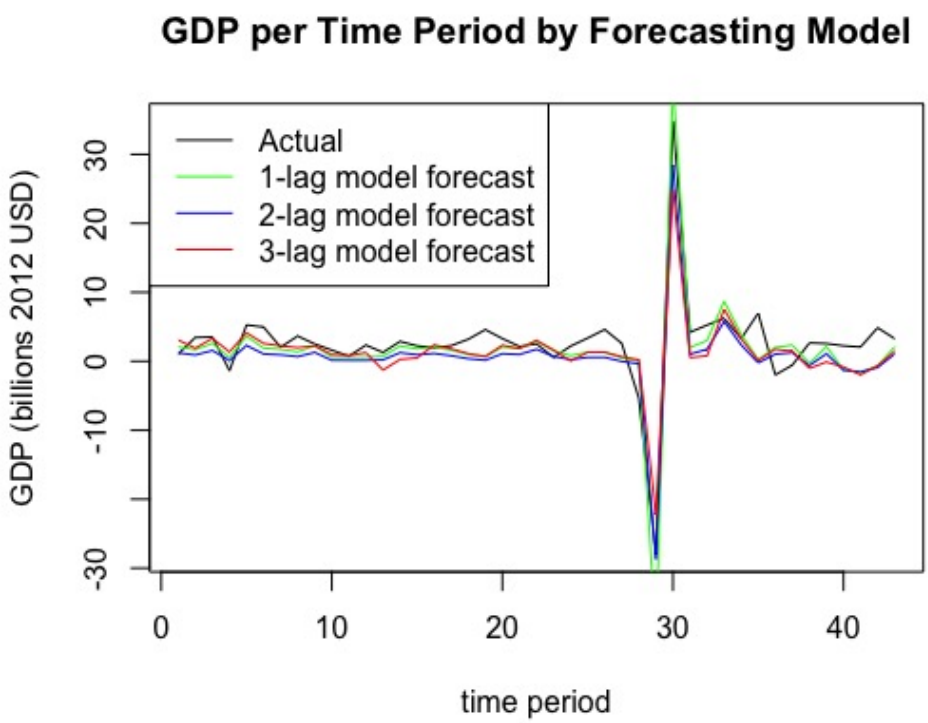
PCA model on lowest frequency data

The PCA model applied to data at the lowest frequency produced the lowest RMSE. First, PCA is applied to the end-of-quarter subset of the 91 monthly variables. Then, the first 11 principal components, which cover nearly 60% of the data variance[1], are used in the model. Regressions are performed for three different lag parameters, all of which identified the coefficient on the first principal component as significant, in addition to the trend and constant coefficients. The first principal component is most correlated with industrial production and number of employees and it is attached to a positive regression coefficient, indicating that these are strong predictors for GDP at least at a quarterly level. The regression model with up to 3 lags per variable also identified the coefficient on the third principal component, which is most correlated with financial metrics such as the bond yield spreads, treasury rates and the S&P index, as significant with a positive regression coefficient. These results agree with some of the observations that were made from an initial exploratory analysis of the data, that industrial production and the S&P 500 index may be good predictors for GDP. Finally, of these three models, the model with variables having up to 2 time lags was the best fit, although it saw only slightly lower RMSE than the other two models. Figure 4.1 shows the validation set actual GDP values as well as the GDP forecasts for these three lag

[1]This threshold was chosen because it covers a reasonable majority of the data variation. No single principal component covers a significant majority of the data variation, as the greatest data variation – explained by the first principal component – is 20% and each principal component after the first adds at most 10% to the total amount of data variation explained. The choice of threshold amount therefore invites a trade-off between number of principal components used in the model and amount of total data variation explained.

models. While the lag 1 and lag 2 models are able to capture the pandemic shocks almost fully, the lag 3 model shows slight underestimates for the extreme swings but marginally closer estimates outside of these swings.

Figure 4.1: Plot of actual GDP versus lower frequency model GDP forecasts during the validation period, 2013-2023



An iteration of this model using only the monthly variables from Schorfheide and Song (2020) suggests that including the full monthly data produces more accurate forecasts. Unlike with the full data, the first 5 principal components are enough to cover 60% of the Schorfheide and Song (2020) data variance and are used in this model. For a 1-lag regression model, the coefficient for the first principal component, which has a strong negative correlation with industrial production, is significant, as well as the trend and constant coefficients. A negative regression coefficient on the first component indicates that its impact on GDP is consistent with results from the PCA model using full monthly data. Indeed, limiting the number of

Table 4.2: This table shows the training and validation RMSEs for the lowest frequency 1-lag PCA model at 3 different number of principal components included.

	Training RMSE	**Validation RMSE**
PCA Model w. 3 PCs (full data)	2.66	2.47
PCA Model w. 11 PCs (full data)	2.66	3.17
PCA Model w. 15 PCs (full data)	2.64	9.39

variables used in this model incorrectly estimates the pandemic GDP swing greatly, leading to an RMSE that is more than double the RMSEs of the full data models. This outcome shows that the monthly variables identified in Schorfheide and Song (2020) alone cannot precisely explain changes in GDP.

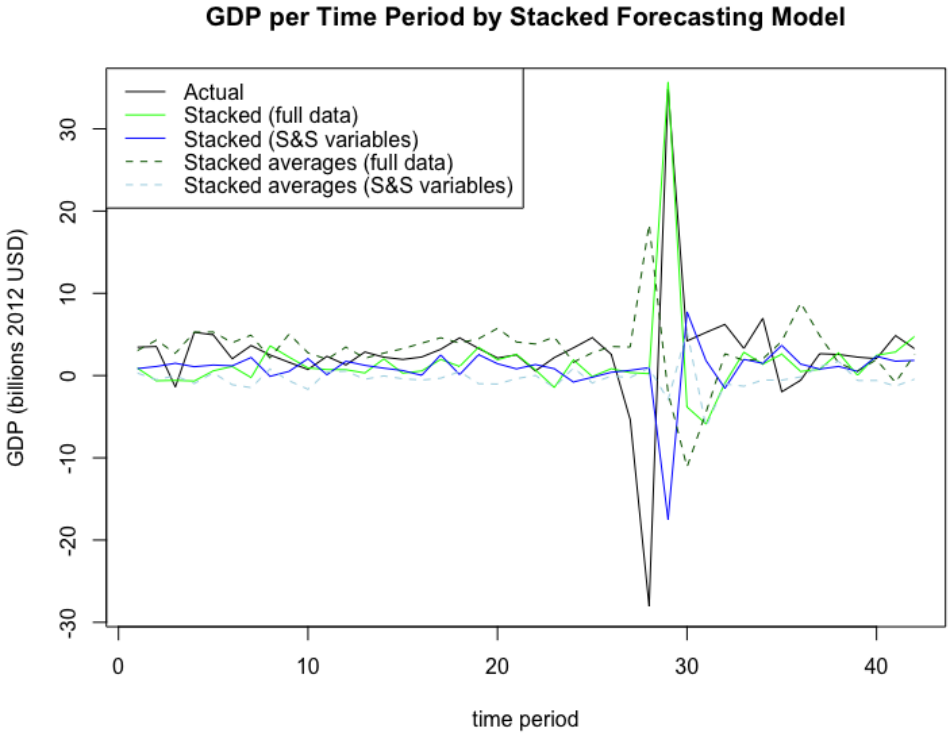
Cross validation can also be used to select the optimal number of principal components to include in the model, instead of relying on a somewhat arbitrary threshold amount of data variance explained. Keeping other parameters constant, adding more principal components to the model using the full data actually increases the RMSE because it overestimates the pandemic swings. The model achieves the lowest error when only three principal components, explaining only 35% of the data variance, are used. The data seems highly sensitive to overfitting, as any additional principal component beyond three can produce exaggerated forecasts for GDP.

PCA model on stacked vector data

The stacked vector data scheme performed worse than the lowest frequency data scheme using the full data. Including all monthly variables into the PCA-transformed data, the variables with significant coefficients in a 1-lag model are the first principal component in months 1 and 2, the month 1 fourth principal component, the month 1 eleventh principal component, GDP from the previous quarter, and a constant. The first component has a negative correlation with industrial production and number of employees mainly, paired

with a negative regression coefficient. The fourth component is positively correlated with treasury rates and has a negative regression coefficient. It is surprising for this component to be identified as significant given that there is no strong relationship between treasury rates and GDP. Lastly, the eleventh principal component is heavily influenced by foreign exchange rates. The GDP forecasts using these regressors misses the drop in GDP in Q2 2020 completely although it fully captures the upswing in Q3 2020. This discrepancy may come from overfitting on the training data, as the training error is less than half the validation error. While the regressors may be able to aptly capture minor GDP changes during stable time periods like in the training data, accurately predicting GDP in the event of external shocks is more dependent on the co-movement of those specific variables with GDP during uncertain times.

Figure 4.2: Plot of actual GDP versus GDP forecasts under the stacked models using full data and Schorfheide and Song (2020) variables during the validation period, 2015-2023



The iteration using only the variables identified in Schorfheide and Song (2020) had a greater training and validation error than the model using the full data, affirming the observation that those specific variables may not be able to fully explain the shifts in GDP. This model also identifies as significant month 1 and 2 of the first principal component, which is mainly related to industrial production, as well as a constant term and a trend term. The increase in validation error comes from forecasting by one quarter later both the Q2 2020 drop in GDP and the Q3 2020 upswing. As with the full data, this discrepancy may depend more on how external shocks uniquely affect the significant regressors. In this case, although monthly industrial production may be a strong predictor of GDP during stable times, during the pandemic, its movements deviated from GDP.

Taking a quarterly average for the monthly variables can help simplify the model by reducing the size of the stacked data matrix. The mean error for the aggregated model increased when using the full data but it decreased when using only the Schorfheide and Song (2020) variables. It is possible that, when including the full data, aggregating the monthly data by quarter smooths out monthly variability that had made GDP forecasts more accurate. On the other hand, stacking the Schorfheide and Song (2020) variables produced GDP forecasts that were off by a quarter, particularly during Q2 and Q3 of 2020, and averaging the monthly data may dampen the signals that increased the forecast error before. The averaged models also identify somewhat different significant variables. For the aggregated model using the full data, the S&P index-correlated fifth component, the S&P dividend yield-correlated sixth component, and the PPI-correlated tenth principal component were identified as significant, in addition to the significant terms from the non-aggregated version. For the aggregated model using only Schorfheide and Song (2020) variables, the CPI-correlated second principal component and GDP are instead identified as the only significant regressors. The mismatch in the latter case suggests that selection of significant regressors for non-aggregated data may have been impacted by high data volatility while estimates are more stable through quarterly averaging.

PCA model on imputed data

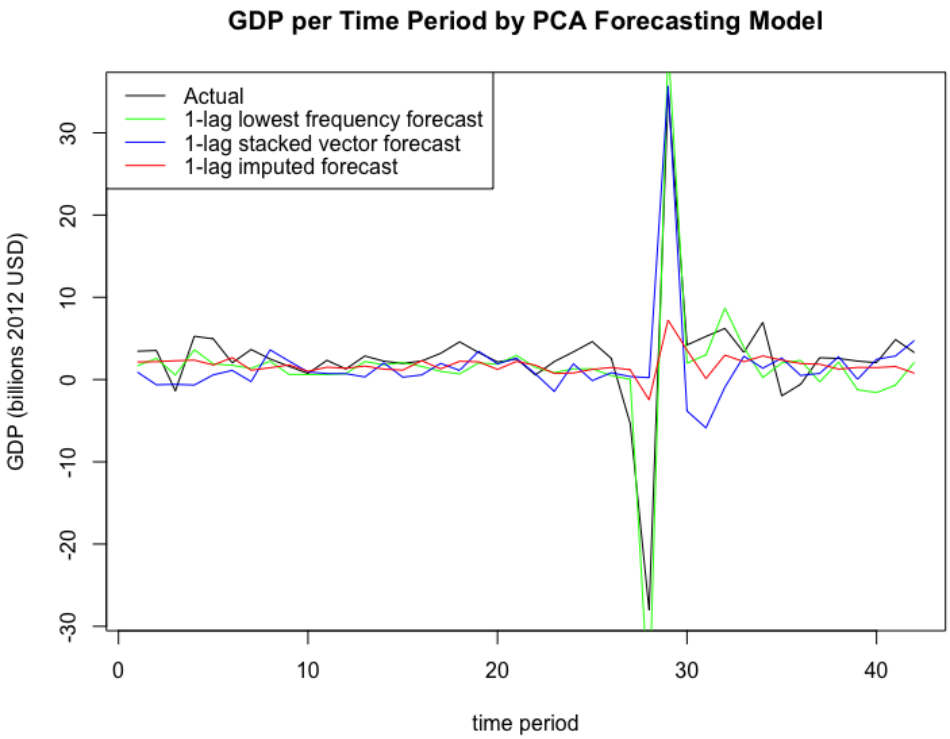
Performance for the imputed data model varied by imputation method, although it generally was worse than that of the best-performing lowest frequency and stacked vector models. The imputation method using exponential smoothing to produce a one-step ahead GDP forecast in the training data performed the worst by far, as expected, given the especially high bias this method introduced. This model identified nearly every principal component as significant and produced extremely volatile GDP forecasts.

Using a cubic spline to impute the training GDP data for the model significantly improved model performance. Applied to the full monthly data, the 1-lag model identified the first, third, and fifth principal components and GDP from the previous month as significant, along with a constant and trend term. Like in the lowest frequency and stacked vector data schemes, the first principal component has a negative correlation with industrial productions and number of employees mainly, along with a negative regression coefficient. The third principal component, attached to a positive regression coefficient, has a positive correlation with treasury rates and bond spreads mainly. Finally, the fifth principal component, also attached to a positive regression coefficient, has a positive correlation with the S&P index mainly. This model wrongly predicts both the direction of change and magnitude of the extreme 2020 Q2 and Q3 GDP swings, showing that greater model bias from imputing GDP increased the forecast error compared to the lowest frequency and stacked vector data schemes but less than the exponentially smoothed GDP imputations did.

Unlike with the other two data schemes, however, limiting the PCA to only the variables from Schorfheide and Song (2020) actually improved model performance. The first principal component and imputed GDP of the previous month, and a constant and trend term, are still identified as the significant regressors. This model directionally estimates the GDP changes accurately and even has a lower mean error on periods outside of the two extreme swings in 2020 than the model using the full data, but the magnitudes of these shocks are severely underestimated. Some of the bias from imputing GDP may have been offset by the

bias from limiting the model to only certain variables, hence why, for only the imputation data scheme, the Schorfheide and Song (2020) variables performed better than the full data.

Figure 4.3: Plot of actual GDP versus GDP forecasts under each data scheme (using the best fit 1-lag model per data scheme) during the validation period, 2015-2023



4.2 Discussion

There are a few learnings that can be gained from these results. First, using GDP by itself cannot produce as accurate a forecast as including other economic variables. Next, industrial production and number of employees are especially important for forecasting GDP. Lastly, using monthly data can make GDP forecasts during time periods without extreme shocks more accurate. But when a rare pandemic affects the economy, the models that were trained on monthly values actually performed worse than a lower frequency model.

Based on the performances of the benchmark models compared to the models using more

than one variable, it is clear that including more variables other than GDP can improve the accuracy of GDP forecasts. This may be the reason that the Federal Reserve Bank considers a whole slew of economic variables when producing their own GDP forecasts. It is also an important assumption for policymakers, given that the policy decisions they make may not directly affect GDP but may affect levers like the unemployment rate which can in turn have implications for GDP. In industry, the forecasts are made using a variety of economic factors as well. Deloitte's (2024) economic outlook for 2025 models a lower growth scenario for GDP in the event of greater tariffs.

In each data scheme, the model identified the principal component relating to industrial production and number of employees as significant, indicating that these variables are especially pertinent to forecasting GDP. Even during an exploratory analysis, industrial production was one of the few variables that saw consistent quarterly co-movement with GDP. It is therefore not surprising that the low-frequency data scheme using this component had the lowest mean error, especially for time periods like the pandemic where GDP saw extreme swings. On the other hand, the monthly changes in industrial production were not as synchronized with the quarterly GDP changes, which explains the fact that the best-fit models in the stacked vector and imputation data schemes performed worse than that of the lowest frequency data scheme. Industrial production recovered in the months immediately following big quarterly shifts, before GDP even saw that shift in the next quarter.

The monthly data appear to more accurately forecast GDP during normal time periods than during periods of extreme shocks, likely from overfitting on relatively stable GDP trends in the training data. The stacked vector and imputation data schemes both saw lower training error than the lowest frequency data scheme. But because the variables had more significant monthly changes during the pandemic, their volatility made it more difficult to accurately predict GDP changes. With the lowest frequency model, including more principal components to cover more data variation also reduced GDP forecast accuracy during abnormal periods although accuracy during the training period is higher.

These observations suggest that it may be productive to have two forecasts – one using monthly data and another using only quarterly data – to get as accurate an estimate for GDP as possible. If the two forecasts are similar, it is likely that the economy is in a stable period and the economic variables are not changing much. On the other hand, the two forecasts being different poses some evidence that there may be external shocks that are first impacting variables outside of GDP. This scenario mirrors the events of the pandemic, in which extreme quarterly movements in industrial production and number of employees preceded massive swings in GDP. Having these points of reference may help policymakers better prepare for extreme shocks.

Chapter 5

Conclusion

In this thesis, I proposed a new method for making more accurate forecasts of U.S. GDP. I reviewed traditional time series models that are often used as benchmarks for new models, then I discussed more recent models that were used to refine GDP forecasts during the pandemic. I subsequently went into greater detail about the FRED dataset I am using, which includes both monthly and quarterly economic variables, and described my principal component regression (PCR) model. To handle the mixed-frequency data, I applied the model across three data schemes. The first scheme uses only the quarterly values of each monthly variable. The second scheme stacks monthly values with quarterly values, and the third scheme imputes monthly values for GDP before fitting a regression model to the data. Including the forecasts from the benchmark models, the PCR model using only quarterly data produced the most accurate forecasts for GDP.

Based on the model performances, there were three main findings from this analysis. First, it is possible to improve GDP forecast accuracy by using economic variables outside of GDP. Next, industrial production and number of employees, particularly at the quarterly level, are good indicators for how GDP will change. Finally, the monthly data produced more accurate GDP forecasts in the absence of unprecedented shocks while the quarterly data saw the opposite performance trend. As a result, it may be productive for American policymakers to compare GDP forecasts from a model using the full monthly data and one using only quarterly data.

There are multiple directions to take future research on this topic. For example, this thesis briefly examined lag assumptions for the model using only quarterly data. However, it is possible that fitting a model with more than one lag can improve model performances for

the stacked and imputed data schemes even more than for the lowest frequency data scheme. In addition, a deeper dive into the relationship between number of principal components used and forecast accuracy may reveal greater insights on the data's sensitivity and potential for overfitting. Parameter regularization may also be included in the model to understand how GDP forecasts may perform under certain limitations, a scenario that is fairly realistic. As these forecasting models are further refined and the statistical tool set grows bigger, U.S. governmental organizations can be more confident in their policy decisions, even during times of the greatest uncertainty.

References

Arruda, J. F. (1992). Surface smoothing and partial spatial derivatives computation using a regressive discrete fourier series. *Mechanical Systems and Signal Processing*, *6*(1), 41-50. Retrieved from https://www.sciencedirect.com/science/article/pii/088832709290055N doi: https://doi.org/10.1016/0888-3270(92)90055-N

Babii, A., Ghysels, E., & Striaukas, J. (2020). *Machine learning time series regressions with an application to nowcasting.* Retrieved from https://arxiv.org/abs/2005.14057

Bai, J., Ghysels, E., & Wright, J. H. (2013, October). State Space Models and MIDAS Regressions. *Econometric Reviews*, *32*(7), 779-813. Retrieved from https://ideas.repec.org/a/taf/emetrv/v32y2013i7p779-813.html doi: 10.1080/07474938.2012.690

Deloitte. (2024). *United states economic forecast q3 2024* (Tech. Rep.). Retrieved from https://www2.deloitte.com/us/en/insights/economy

Foroni, C., Marcellino, M., & Stevanovic, D. (2020, June). *Forecasting the COVID-19 recession and recovery: Lessons from the financial crisis* (Working Papers No. 20-14). Chair in macroeconomics and forecasting, University of Quebec in Montreal's School of Management. Retrieved from https://ideas.repec.org/p/bbh/wpaper/20-14.html

Gardner, E. S. (2006). Exponential smoothing: The state of the art—part ii. *International Journal of Forecasting*, *22*(4), 637-666. Retrieved from https://www.sciencedirect.com/science/article/pii/S0169207006000392 doi: https://doi.org/10.1016/j.ijforecast.2006.03.005

Gardner, E. S., & McKenzie, E. (1985). Forecasting trends in time series. *Management Science*, *31*(10), 1237–1246. Retrieved 2024-12-05, from http://www.jstor.org/stable/2631713

Ghysels, E. (2016). Macroeconomics and the reality of mixed frequency data. *Journal of Econometrics*, *193*(2), 294-314. Retrieved from https://www.sciencedirect.com/science/article/pii/S0304407616300653 (The Econometric Analysis of Mixed Frequency Data Sampling)

Ghysels, E., Santa-Clara, P., & Valkanov, R. (2004, May). *The MIDAS Touch: Mixed Data Sampling Regression Models* (CIRANO Working Papers No. 2004s-20). CIRANO. Retrieved from https://ideas.repec.org/p/cir/cirwor/2004s-20.html

Holt, C. C. (2004). Forecasting seasonals and trends by exponentially weighted moving averages. *International Journal of Forecasting*, *20*(1), 5-10. Retrieved from https://www.sciencedirect.com/science/article/pii/S0169207003001134 doi: https://doi.org/10.1016/j.ijforecast.2003.09.015

Hyndman, R., & Khandakar, Y. (2008, 07). Automatic time series forecasting: The forecast package for r. *Journal of Statistical Software*, *26*. doi: 10.18637/jss.v027.i03

Hyndman, R. J., Koehler, A. B., Snyder, R. D., & Grose, S. (2002). A state space framework for automatic forecasting using exponential smoothing methods. *International Journal of Forecasting*, *18*(3), 439-454. Retrieved from https://www.sciencedirect.com/science/article/pii/S0169207001001108 doi: https://doi.org/10.1016/S0169-2070(01)00110-8

Landi, D. M., & Johnson, B. M. (1967). Optimal linear inventory control and first order exponential smoothing. *Operations Research*, *15*(2), 342-347. Retrieved from https://doi.org/10.1287/opre.15.2.342

Makridakis, S., & Hibon, M. (1991). Exponential smoothing: The effect of initial values and loss functions on post-sample forecasting accuracy. *International Journal of Forecasting*, *7*(3), 317-330. Retrieved from https://www.sciencedirect.com/science/article/pii/016920709190005G doi: https://doi.org/10.1016/0169-2070(91)90005-G

McCracken, M. W., & Ng, S. (2021, January). FRED-QD: A Quarterly Database for

Macroeconomic Research. *Review*, *103*(1), 1-44. doi: 10.20955/r.103.1-44

Mélard, G., & Pasteels, J.-M. (2000). Automatic arima modeling including interventions, using time series expert software. *International Journal of Forecasting*, *16*(4), 497-508. Retrieved from https://www.sciencedirect.com/science/article/pii/S0169207000000674 (The M3- Competition) doi: https://doi.org/10.1016/S0169-2070(00)00067-4

Newbold, P., & Bos, T. (1989). On exponential smoothing and the assumption of deterministic trend plus white noise data-generating models. *International Journal of Forecasting*, *5*(4), 523-527. Retrieved from https://www.sciencedirect.com/science/article/pii/0169207089900071 doi: https://doi.org/10.1016/0169-2070(89)90007-1

Schorfheide, F., & Song, D. (2013, December). *Real-time forecasting with a mixed-frequency var* (Working Paper No. 19712). National Bureau of Economic Research. Retrieved from http://www.nber.org/papers/w19712 doi: 10.3386/w19712

Schorfheide, F., & Song, D. (2021, December). *Real-time forecasting with a (standard) mixed-frequency var during a pandemic* (Working Paper No. 29535). National Bureau of Economic Research. Retrieved from http://www.nber.org/papers/w29535 doi: 10.3386/w29535

Trigg, D., & Pitts, E. (1962). The optimal choice of the smoothing constant in an inventory control system. *OR*, *13*(4), 287–298. Retrieved 2024-12-05, from http://www.jstor.org/stable/3007036

Appendix

Taken from Hyndman et al. (2002), this table shows the component formulations for the ETS model.

Formulae for recursive calculations and point forecasts

Trend component	Seasonal component		
	N (none)	A (additive)	M (multiplicative)
N (none)	$P_t = Y_t$ $Q_t = l_{t-1}$ $\phi = 1$ $F_{t+h} = l_t$	$P_t = Y_t - s_{t-m}$ $Q_t = l_{t-1}$ $T_t = Y_t - Q_t$ $\phi = 1$ $F_{t+h} = l_t + s_{t+h-m}$	$P_t = Y_t / s_{t-m}$ $Q_t = l_{t-1}$ $T_t = Y_t / Q_t$ $\phi = 1$ $F_{t+h} = l_t s_{t+h-m}$
A (additive)	$P_t = Y_t$ $Q_t = l_{t-1} + b_{t-1}$ $R_t = l_t - l_{t-1}$ $\phi = 1$ $F_{t+h} = l_t + hb_t$	$P_t = Y_t - s_{t-m}$ $Q_t = l_{t-1} + b_{t-1}$ $R_t = l_t - l_{t-1}$ $T_t = Y_t - Q_t$ $\phi = 1$ $F_{t+h} = l_t + hb_t + s_{t+h-m}$	$P_t = Y_t / s_{t-m}$ $Q_t = l_{t-1} + b_{t-1}$ $R_t = l_t - l_{t-1}$ $T_t = Y_t / Q_t$ $\phi = 1$ $F_{t+h} = (l_t + hb_t) s_{t+h-m}$
M (multiplicative)	$P_t = Y_t$ $Q_t = l_{t-1} b_{t-1}$ $R_t = l_t / l_{t-1}$ $\phi = 1$ $F_{t+h} = l_t b_t^h$	$P_t = Y_t - s_{t-m}$ $Q_t = l_{t-1} b_{t-1}$ $R_t = l_t / l_{t-1}$ $T_t = Y_t - Q_t$ $\phi = 1$ $F_{t+h} = l_t b_t^h + s_{t+h-m}$	$P_t = Y_t / s_{t-m}$ $Q_t = l_{t-1} b_{t-1}$ $R_t = l_t / l_{t-1}$ $T_t = Y_t / Q_t$ $\phi = 1$ $F_{t+h} = l_t b_t^h s_{t+h-m}$
D (damped)	$P_t = Y_t$ $Q_t = l_{t-1} + b_{t-1}$ $R_t = l_t - l_{t-1}$ $\beta < \phi < 1$ $F_{t+h} = l_t +$ $(1 + \phi + \cdots + \phi^{h-1}) b_t$	$P_t = Y_t - s_{t-m}$ $Q_t = l_{t-1} + b_{t-1}$ $R_t = l_t - l_{t-1}$ $T_t = Y_t - Q_t$ $\beta < \phi < 1$ $F_{t+h} = l_t +$ $(1 + \phi + \cdots + \phi^{h-1}) b_t + s_{t+h-m}$	$P_t = Y_t / s_{t-m}$ $Q_t = l_{t-1} + b_{t-1}$ $R_t = l_t - l_{t-1}$ $T_t = Y_t / Q_t$ $\beta < \phi < 1$ $F_{t+h} = [l_t +$ $(1 + \phi + \cdots + \phi^{h-1}) b_t] s_{t+h-m}$